

RIGIDITY AND UNIVERSAL CONTRACTIONS OF HARD-SPHERE KINETIC COTANGENTS

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ABSTRACT. We derive full path-space LDPs from compatible finite-dimensional pressures which are finite, differentiable, and steep on their support quotients, followed by Dawson–Gärtner passage and exponential tightness. The resulting global rate uses only the proved cylinder source algebra and requires no rate-dense exposed-point hypothesis. Typed contraction, exact source-dependent conditioning, their closed joint-infimum interchange, and additive dynamic programming commute with pressure derivatives and Gaussian tangents. Every local canonical likelihood is exactly centred at the finite-volume mean and is a mean-one Radon–Nikodym density.

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1. ROUND-SIX POSITIVE CLOSURE: PROJECTIVE STEEP PRESSURES AND DETERMINISTIC COMMUTATION

The global rate is derived from full finite-dimensional pressure theorems on an increasing separating source algebra. No local analytic ball is evaluated outside its domain, and no rate-dense exposed-point lower bound is assumed. The finite-dimensional lower bounds follow from global differentiability and steepness of each projected pressure; exponential tightness then supplies the projective full LDP. Every local likelihood is centred at the exact finite-volume mean.

1.1. A separating projective source system. Let E be a Hausdorff path state space and let

$$\pi_m : E \longrightarrow \mathbb{R}^{d_m}, \quad m \geq 1,$$

be continuous maps such that $\pi_m = R_{m,n}\pi_n$ for $m < n$, and the family separates points and generates the topology on compact sets. Let X_ε be E -valued and let $\mu_\varepsilon \rightarrow \infty$. Define

$$Q_{\varepsilon,m}(\theta) = \frac{1}{\mu_\varepsilon} \log \mathbb{E} \exp\{\mu_\varepsilon \theta \cdot \pi_m(X_\varepsilon)\}.$$

The hypotheses are:

- (P1) for every m , $Q_{\varepsilon,m} \rightarrow Q_m$ locally uniformly on $\mathbb{C}^{d_m}$, and all finite pressures are locally uniformly bounded on complex compact sets;
- (P2) Q_m is finite, differentiable on all of $\mathbb{R}^{d_m}$, and steep in the sense that $|\nabla Q_m(\theta_n)| \rightarrow \infty$ whenever $|\theta_n| \rightarrow \infty$ along a direction not annihilated by the support;
- (P3) the pressures are projectively compatible: $Q_m(\theta) = Q_n(R_{m,n}^T \theta)$;
- (P4) (X_ε) is exponentially tight in E .

For a platform with conservation directions, the statements are read on the affine support quotient, where steepness is required.

Proposition 1.1 (Exact finite-volume normalization). *For directions $h, k \in \mathbb{R}^{d_m}$,*

$$DQ_{\varepsilon,m}(\theta)h = \mathbb{E}_{\theta,\varepsilon}[h \cdot \pi_m(X_\varepsilon)],$$

$$D^2Q_{\varepsilon,m}(\theta)[h, k] = \mu_\varepsilon \text{Cov}_{\theta,\varepsilon}(h \cdot \pi_m(X_\varepsilon), k \cdot \pi_m(X_\varepsilon)).$$

All derivatives converge locally uniformly to those of Q_m .

Proof. Differentiate the finite logarithmic moment generating function. One factor of μ_ε remains in the second derivative after the outer normalization. Normal-family convergence and Cauchy's formula give convergence of every derivative on smaller complex compact sets. $\square$

1.2. The projective full LDP without an exposed-density hypothesis.

Put

$$I_m(y) = \sup_{\theta \in \mathbb{R}^{d_m}} \{\theta \cdot y - Q_m(\theta) + Q_m(0)\}$$

and

$$I(x) = \sup_m I_m(\pi_m x).$$

Theorem 1.2 (Projective steep-pressure LDP). *Under (P1)–(P4), (X_ε) satisfies a full LDP on E with good rate I . Equivalently, if $\mathcal{D} = \bigcup_m \pi_m^*(\mathbb{R}^{d_m})$ is the proved cylinder source algebra, then*

$$I(x) = \sup_{F \in \mathcal{D}} \{F(x) - Q(F) + Q(0)\}.$$

The theorem does not use a pressure outside $\mathcal{D}$ and does not assume rate-density of exposed points.

Proof. For fixed m , the finite-dimensional Gärtner–Ellis theorem applies because Q_m is finite, differentiable, and steep on the support quotient. It gives a full LDP for $\pi_m(X_\varepsilon)$ with good rate I_m , including the lower bound at every point of its effective domain. Projective compatibility makes these finite-dimensional rates consistent. The Dawson–Gärtner theorem gives the projective-limit LDP with rate $\sup_m I_m$ in the initial projective topology. Exponential tightness in E upgrades the weak projective upper bound to the full topology and makes the rate good. Rewriting the supremum over (m, θ) gives the cylinder source formula. $\square$

For the Sinai chain, A2 supplies the global finite-cylinder twisted pressures and A3 supplies exponential tightness. For the hard-sphere chain, B2 source sewing supplies every finite bounded real cylinder source, B3 supplies strict quotient differentiability, and B2 compact containment supplies (P4).

1.3. Typed contractions. Let $T : E \rightarrow F$ be continuous, or have an exponentially good sequence of continuous approximants. Let its finite coordinates factor through the projective system.

Theorem 1.3 (Pressure/rate contraction commutation). *The contracted variables satisfy a full LDP with*

$$I_T(y) = \inf_{Tx=y} I(x).$$

For every finite source η on F ,

$$Q_T(\eta) = Q(T^*\eta).$$

On a local analytic chart,

$$DQ_T = T DQ, \quad D^2Q_T = T(D^2Q)T^*.$$

The same statements hold for exponentially good approximations by the extended contraction theorem.

Proof. The rate formula is the contraction principle applied to Theorem 1.2. The pressure identity holds exactly at finite volume because $\langle \eta, TX \rangle = \langle T^* \eta, X \rangle$. Differentiate the local analytic identity. Exponentially good approximants have the same pressure and rate limits by exponential equivalence. $\square$

1.4. Source-dependent conditioning. Let $C : E \rightarrow \mathbb{R}^k$ be a continuous finite-coordinate constraint and let $a_\varepsilon \rightarrow a$. Assume the B1-type exact coefficient property: for every bounded cylinder source F , there is a finite multiplier $\lambda_{F,\varepsilon}$ satisfying

$$D_\lambda Q_\varepsilon(F + C^* \lambda_{F,\varepsilon}) = a_\varepsilon$$

and the corresponding shell has logarithmic probability $o(\mu_\varepsilon)$. Assume the constraint Hessian is positive on the support quotient.

Define

$$Q^a(F) = \inf_\lambda \{Q(F + C^* \lambda) - \lambda \cdot a\} - \inf_\lambda \{Q(C^* \lambda) - \lambda \cdot a\}.$$

Theorem 1.4 (Exact conditioning commutation). *The microscopic conditional laws satisfy a full LDP with good rate*

$$I^a(x) = \begin{cases} I(x) - \inf_{Cy=a} I(y), & Cx = a, \\ +\infty, & Cx \neq a. \end{cases}$$

Their cylinder pressure is Q^a . On a regular source chart the unique limiting multiplier is source dependent and

$$D^2 Q^a = Q_{FF} - Q_{F\lambda} Q_{\lambda\lambda}^{-1} Q_{\lambda F}.$$

Proof. Change measure at the exact finite-volume multiplier. The coefficient hypothesis makes the shell probability subexponential under that law, so the conditional cylinder pressure is the displayed finite-dimensional infimum. The finite projected conditional laws therefore satisfy Gärtner–Ellis with that pressure. Projective passage and conditional exponential tightness give the full LDP. Alternatively, the rate follows by restricting the good unconditioned rate to the closed constraint surface and subtracting its minimum. Strict constraint convexity gives the unique multiplier. Differentiating the saddle equation and then the envelope gives the Schur complement. $\square$

1.5. Conditioning and contraction in either order. Let $T : E \rightarrow F$ and $C : E \rightarrow \mathbb{R}^k$ be jointly continuous finite coordinate maps.

Theorem 1.5 (Closed joint-infimum interchange). *The two routes*

$$E \xrightarrow{\text{condition } C=a} E_a \xrightarrow{T} F$$

and

$$E \xrightarrow{(T,C)} F \times \mathbb{R}^k \xrightarrow{\text{condition second coordinate } a} F$$

have the same good rate

$$I_{T|a}(y) = \inf_{Tx=y, Cx=a} I(x) - \inf_{Cx=a} I(x).$$

On every regular finite-dimensional source chart this rate is the conjugate of $Q^a(T^\eta)$. No duality gap occurs.*

Proof. Apply the contraction principle to the joint map (T, C) and then restrict to the closed slice with second coordinate a . Since I is good, every finite joint infimum is attained and infimal projection is lower semicontinuous on its effective domain. This proves equality of the two primal routes without a minimax interchange. On a regular finite-dimensional chart, the joint projected pressure is finite, differentiable, and steep; its Legendre transform is the joint rate. Eliminating the constraint multiplier by the strictly convex saddle gives the conjugate of Q^a , hence no gap. $\square$

1.6. Exact local likelihoods and Gaussian tangents. Fix a projected source $\theta \in \mathbb{R}^{d_m}$. Put

$$m_{\varepsilon, \theta} = DQ_{\varepsilon, m}(\theta), \quad Z_{\varepsilon, \theta} = \sqrt{\mu_\varepsilon}(\pi_m(X_\varepsilon) - m_{\varepsilon, \theta}).$$

For $h \in \mathbb{R}^{d_m}$, define

$$L_\varepsilon(h) = \exp \left\{ h \cdot Z_{\varepsilon, \theta} - \mu_\varepsilon [Q_{\varepsilon, m}(\theta + h/\sqrt{\mu_\varepsilon}) - Q_{\varepsilon, m}(\theta) - \mu_\varepsilon^{-1/2} DQ_{\varepsilon, m}(\theta)h] \right\}.$$

Theorem 1.6 (Exact finite-centred likelihood and Gaussian shift). *For every finite ε , $L_\varepsilon(h)$ is exactly the Radon–Nikodym density between the sources $\theta + h/\sqrt{\mu_\varepsilon}$ and θ ; hence it has mean one. Jointly,*

$$(Z_{\varepsilon, \theta}, L_\varepsilon(h)) \implies (Z, e^{h \cdot Z - \frac{1}{2} h^T \Sigma_\theta h}), \quad \Sigma_\theta = D^2 Q_m(\theta).$$

Under the tilted likelihood the limit mean is shifted by $\Sigma_\theta h$. After conditioning, replace Σ_θ by its constraint Schur complement; after contraction, replace it by $T\Sigma_\theta T^$.*

Proof. Substitute the definition of $Z_{\varepsilon, \theta}$; the finite mean terms cancel algebraically and leave

$$\sqrt{\mu_\varepsilon} h \cdot \pi_m(X_\varepsilon) - \mu_\varepsilon [Q_{\varepsilon, m}(\theta + h/\sqrt{\mu_\varepsilon}) - Q_{\varepsilon, m}(\theta)],$$

the exact likelihood exponent. Taylor expansion on a larger complex compact and Proposition 1.1 give the Gaussian characteristic function and likelihood limit. A slightly larger real source ball gives an $L^{1+\delta}$ bound for fixed h , hence uniform integrability. Le Cam’s third lemma gives the shift. The conditioning and contraction formulas follow from Theorems 1.4 and 1.3. $\square$

A process-valued Gaussian conclusion additionally requires tightness. In the hard-sphere platform it is supplied by the B3 localized-cumulant theorem; in the Sinai platform it is supplied by the A4 enhanced conditional theorem. The finite-dimensional analytic theorem above does not silently claim that tightness.

1.7. The dynamic rate–pressure–semigroup triangle. Assume a path rate has the additive form

$$I_{r,t}(x) = I_{r,s}(x|_{[r,s]}) + I_{s,t}(x|_{[s,t]})$$

on compatible concatenations and compact sublevels. Define

$$(S_{r,t}\Phi)(x_r) = \sup_{x:x(r)=x_r} \{\Phi(x_t) - I_{r,t}(x)\}.$$

Theorem 1.7 (Dynamic commutation theorem). *The additive rate generates a nonlinear semigroup $S_{r,t} = S_{r,s}S_{s,t}$. On every regular cylinder chart its infinitesimal Hamiltonian is the pressure derivative dual to the local action. Quadratic expansion of the rate, pressure, and semigroup around an exposed phase gives the same Gaussian covariance and the exponential-transform generator*

$$\mathfrak{G}^\vartheta F = b \cdot DF + \frac{1}{2}A : D^2F + \frac{\vartheta}{2}DF^TADF.$$

Typed conditioning and contraction may be performed before or after this quadratic tangent according to the Schur-complement and covariance-pushforward rules above.

Proof. Split any admissible path at time s . Additivity gives one inequality in the semigroup identity; concatenating near optimizers gives the reverse. Fenchel duality on a short interval identifies the infinitesimal Hamiltonian. Taylor expansion of the compatible finite-dimensional pressures yields the Gaussian quadratic form, whose Legendre transform is the quadratic tangent of the action. Applying the logarithmic exponential transform to the Gaussian generator adds the final quadratic-gradient term. The finite-dimensional matrix identities for conditioning and contraction commute with this expansion. $\square$

1.8. Conditional calibration. A smooth cash-additive and strongly time-consistent utility-generated certainty equivalent is affine or entropic with a constant coefficient. That coefficient equals a mechanical preparation field only when the two normalized likelihood cocycles are required to coincide in the same units.

Theorem 1.8 (Round-six D1 closure). *For the proved projective source systems of the Sinai and hard-sphere platforms, full LDP formation commutes with typed contraction, exact source-dependent conditioning, finite-dimensional pressure differentiation, Gaussian likelihood formation, and additive dynamic programming. The global rate is derived from finite-dimensional steep pressures and exponential tightness rather than an assumed exposed-density theorem, and every finite likelihood is exactly centred at DQ_ε .*

Proof. The global LDP is Theorem 1.2. Contraction, conditioning, and their interchange are Theorems 1.3, 1.4, and 1.5. Exact likelihoods are Theorem 1.6, and dynamic commutation is Theorem 1.7. The calibration qualification was stated in the preceding subsection. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-three dependency ledger. Clean compilation and automated structural checks are necessary verification gates; independent external journal review remains a separate process.

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